Student Performance analytics Dashboard Azure Concepts and Power BI

A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled "Student Performance Analytics Dashboard using Azure Concepts and Power BI" is a record of bonafide work carried out by team members **Pawan Achyutanand – 2320030053, Ponnam Harshavardhan – 2320030011, K Pranay – 2320030481, and Paladi Abhideep – 2320030175**, submitted in partial fulfilment for the award of B.Tech in Computer Science and Engineering to K L University. The results embodied in this report have not been copied from any other department, university, or institute and are the original work of the team.

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Certificate

This is to certify that the project-based report entitled "**Student Performance Analytics Dashboard using Azure Concepts and Power BI**" is a bonafide work carried out and submitted by **Pawan Achyutanand – 2320030053, Ponnam Harshavardhan – 2320030011, K Pranay – 2320030481, and Paladi Abhideep – 2320030175**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

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Signature of the External Examiner

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, **Prathusha Bonagani**, Department of Computer Science which cannot be expressed in words for his tremendous support, encouragement and guidance for this project.

We express our gratitude to Dr.N. Sirisha, Head of the Department for Computer Science Engineering for providing us with adequate facilities, ways and means by which we can complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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ABSTRACT

In recent years, the rapid growth of digital education systems has generated a massive amount of academic data, making traditional data analysis methods inefficient and time-consuming. This project, titled "Student Performance Analytics Dashboard using Azure Concepts and Power BI", aims to analyze student academic data to identify performance trends, subject-wise strengths and weaknesses, and the impact of attendance on overall results.

The project utilizes Microsoft Azure SQL Database for secure and scalable data storage. Student data, including marks, attendance, and subject details, is stored in a structured relational format and directly connected to Power BI for analysis and visualization. The dataset consists of key attributes such as student information, subject scores, attendance records, and performance metrics.

Data preprocessing and analysis are performed within Power BI using DAX (Data Analysis Expressions), where various analytical measures are created to compute key performance indicators such as average marks, pass percentage, fail count, and subject-wise performance trends. Visualization techniques such as bar charts, pie charts, and KPI cards are used to present insights in an interactive and user-friendly dashboard.

The results demonstrate that cloud-based analytics enables efficient processing and visualization of academic data, providing valuable insights for educators to monitor student progress and improve learning outcomes. This project highlights the importance of cloud computing and data analytics in modern education systems and shows how scalable platforms can support better decision-making, performance tracking, and future enhancements like real-time analytics and predictive modeling.

INTRODUCTION

In today's digital era, educational institutions generate a massive volume of academic data through student records, examination results, and attendance systems. Managing and analysing such large-scale data using traditional manual methods has become increasingly difficult, time-consuming, and error-prone. This has led to the adoption of cloud-based data analytics solutions, which provide scalable, efficient, and cost-effective ways to handle and analyse academic datasets.

The project titled "Student Performance Analytics Dashboard using Azure Concepts and Power BI" focuses on analysing student academic data using modern cloud and visualization technologies. The main objective of this project is to study student performance trends, evaluate subject-wise outcomes, and understand the impact of attendance on academic results through an interactive dashboard.

In this project, Microsoft Azure SQL Database is used as the cloud platform for storing structured student data securely and efficiently. The stored data is then connected to Power BI, which is used for data processing, analysis, and visualization. By leveraging cloud computing and business intelligence tools, the project demonstrates how educational institutions can derive meaningful insights from academic data in a faster, more reliable, and user-friendly manner.

Problem Description and Objectives:

Educational institutions handle large volumes of student data, including marks, attendance, and subject performance across multiple semesters. Analysing this data using traditional methods such as spreadsheets or manual calculations is inefficient, time-consuming, and prone to errors. As the complexity of data increases, it becomes difficult to identify performance trends, detect weak subjects, and evaluate overall academic progress. These limitations make it challenging for educators to make timely and informed decisions. To address this issue, cloud-based data analytics provides a scalable and efficient solution with secure storage, high computational power, and faster processing capabilities. This project implements a student performance analytics system using Microsoft Azure SQL Database and Power BI to efficiently manage and analyze academic data.

The main objective of this project is to analyze student performance data and extract meaningful insights to support academic decision-making. The system stores structured data in Azure SQL Database and performs analysis using Power BI to identify trends such as subject-wise performance, average marks, and pass/fail distribution. It also evaluates the relationship between attendance and academic outcomes while generating key performance indicators (KPIs) like total students, average marks, pass percentage, and fail count. Interactive dashboards and visualizations are used to present insights clearly and effectively. Overall, the project demonstrates how cloud computing and data analytics can enhance performance tracking, improve decision-making, and increase efficiency in educational institutions.

LITERATURE SURVEY

The rapid growth of digital education systems, online learning platforms, and academic management systems has led to a significant increase in educational data. Traditional data processing techniques are no longer sufficient to handle such large-scale and continuously growing datasets. As a result, cloud computing and data analytics have emerged as essential technologies for analysing academic data efficiently. Researchers have explored the use of cloud platforms, data analytics, and visualization techniques to improve student performance monitoring, academic decision-making, and overall institutional efficiency.

Student performance analytics has become an important research area due to its role in identifying learning patterns, evaluating subject-wise performance, and improving educational outcomes. With the adoption of cloud infrastructure, institutions can store large volumes of academic data and perform complex analysis in a scalable and cost-effective manner. Cloud platforms such as Microsoft Azure, Amazon Web Services, and Google Cloud Platform have significantly transformed traditional academic data processing into modern, data-driven systems.

Cloud Computing in Educational Data Processing

Cloud computing has revolutionized the way academic data is stored and processed. According to Armbrust et al. (2010), cloud platforms provide scalable computing resources that allow institutions to manage large datasets without investing in physical infrastructure. Educational organizations benefit from cloud environments due to high availability, scalability, and reliability.

Microsoft Azure SQL Database is widely used for storing structured academic data due to its security, scalability, and seamless integration with analytics tools. Studies show that cloud-based storage enables faster data access and efficient data management, making it suitable for academic systems where data consistency and accessibility are critical.

Data Analytics in Educational Systems

Data analytics plays a vital role in extracting meaningful insights from large volumes of student data. Research by Chen et al. (2012) highlights that analytics techniques help institutions identify performance trends, student behaviour, and learning patterns. Analytical tools enable institutions to evaluate academic performance and take corrective actions.

In this context, Power BI is widely used for data analysis and visualization. It supports advanced analytical capabilities through DAX (Data Analysis Expressions), enabling the calculation of key performance indicators (KPIs) such as average marks, pass percentage, and performance trends. Compared to traditional systems, modern analytics tools provide faster insights and improved decision-making capabilities.

Student Performance Analysis

Student performance analysis focuses on evaluating academic outcomes based on marks, attendance, and subject-wise results. Researchers have shown that identifying patterns such as low performance in specific subjects or irregular attendance can help institutions provide targeted support to students.

Various analytical approaches, including statistical analysis and rule-based evaluation, are used to assess performance. For example, students scoring below a certain threshold can be identified as needing improvement. Unlike financial fraud detection, academic analytics focuses more on performance evaluation, trend analysis, and outcome improvement rather than anomaly detection.

Visualization and Decision Support Systems

Data visualization plays a crucial role in educational analytics by converting complex datasets into understandable graphical formats. According to Few (2009), visual analytics improves the understanding of patterns, trends, and relationships within data.

Tools like Power BI enable the creation of interactive dashboards using bar charts, pie charts, KPI cards, and trend graphs. These visualizations help educators quickly analyse student performance, compare subject-wise results, and monitor progress. Interactive dashboards enhance decision-making by presenting insights in a clear and intuitive manner.

Comparative Studies and Observations

Several studies comparing traditional data processing methods with cloud-based analytics have concluded that cloud platforms offer better scalability, performance, and reliability. While traditional systems struggle with large datasets, cloud-based solutions efficiently handle high data volume and complexity.

Research also indicates that combining cloud storage with visualization tools provides an effective solution for academic analytics without requiring complex machine learning models. The integration of Microsoft Azure SQL Database and Power BI enables efficient data storage, processing, and Visualization, making it ideal for student performance analysis.

Conclusion

The literature clearly demonstrates that cloud-based analytics has become essential for modern educational systems. From traditional data analysis methods to advanced cloud-powered analytics and visualization tools, research highlights the advantages of scalable computing and intelligent data processing.

This project builds upon these research findings by implementing a cloud-based student performance analytics system using Microsoft Azure SQL Database and Power BI. By analyzing student data, identifying performance trends, and visualizing insights through interactive dashboards, the project demonstrates the practical application of cloud computing in education and its effectiveness in improving academic performance tracking and decision-making.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Specifications

The Cloud-Based Financial Transaction Pattern Analytics project does not require high-end local hardware, as most of the processing is performed in the cloud using Microsoft Azure. However, a basic system configuration is required to access cloud services and perform analysis smoothly.

Processor

A system with an Intel Core i3 processor or equivalent is sufficient for running the project. For better performance and smooth multitasking, an Intel Core i5 or higher is recommended. Since the data processing is handled by Azure Databricks, the local processor is mainly used for accessing the platform and visualizing results.

RAM

A minimum of 4 GB RAM is required to run the browser, development tools, and cloud interfaces efficiently. However, 8 GB RAM or above is recommended to ensure better performance while running Databricks notebooks and handling multiple tabs simultaneously.

Storage

The project does not consume large local storage since the dataset is stored in Azure Blob Storage. However, at least 500 MB of free disk space is required for:

- Browser cache
- Temporary files
- Python-related dependencies

Graphics

No dedicated graphics card is required. Integrated graphics available in modern processors are sufficient, as the project focuses on data processing and visualization rather than graphical rendering.

Operating System

The project is compatible with:

- Windows
- Linux
- macOS

Since the system is cloud-based, the operating system does not affect the execution of the project.

Software Requirements

The project uses cloud-based tools and open-source technologies to perform data analytics efficiently.

Cloud Platform

- Microsoft Azure
  - Azure Blob Storage – for storing financial transaction data
  - Azure Databricks – for data processing and analytics

Programming Language

- Python (Version 3.7 or above)

  Python is used for data processing, analytics, and machine learning operations in Azure Databricks.

Libraries and Frameworks

The following Python libraries are used in the project:

- Apache Spark (PySpark) – For distributed data processing
- Pandas – For data manipulation
- NumPy – For numerical computations
- Matplotlib / Seaborn (optional) – For visualization
- PySpark MLlib – For basic machine learning and fraud detection

These libraries are either pre-installed in Azure Databricks or easily available within the environment.

Development Environment

- Azure Databricks Workspace – Main development and execution platform
- Web Browser (Google Chrome / Microsoft Edge) – To access Azure Portal
- Visual Studio Code (Optional) – For local code editing or documentation

Visualization Tools

- Databricks Built-in Visualization Tools
    - Bar charts
    - Pie charts
    - Tables

These tools are used to represent transaction patterns, customer behaviour, and fraud analysis results.

With the above hardware and software configuration, the Cloud-Based Financial Transaction Analytics system can be executed efficiently. Since the majority of computation is performed in the cloud, the project does not require high-end local hardware. The use of Azure Databricks ensures scalability, performance, and reliability, making the system suitable for analysing large financial datasets and identifying transaction patterns effectively.

IMPLEMENTATION

Implementation of Student Performance Analytics Dashboard

The implementation of the *Student Performance Analytics Dashboard* focuses on analysing academic data using cloud-based technologies. The system is designed to store, process, analyse, and visualize student performance data efficiently using Microsoft Azure SQL Database and Power BI. The complete implementation involves structured data storage, data processing, analytical computation, and interactive visualization.

The implementation process is divided into multiple stages, including data storage, data preprocessing, analytical computation, and dashboard visualization. Each stage plays a crucial role in ensuring accurate analysis and efficient handling of academic data.

1. Data Representation and Dataset Description

The dataset used in this project consists of student academic records. Each record represents a student's performance and includes various attributes related to academics.

The dataset includes the following key attributes:

- Student ID – Unique identifier for each student
- Student Name – Name of the student
- Subject – Course or subject name
- Marks – Score obtained by the student
- Attendance – Attendance percentage
- Grade – Performance grade
- Semester – Academic term

The dataset is stored in Azure SQL Database in a structured relational format using tables such as Students, Subjects, Marks, and Attendance.

2. Establishing the Cloud Environment

The implementation environment is created using Microsoft Azure services and Power BI.

Azure SQL Database

- Used for storing structured student data
- Supports relational schema with tables and relationships
- Provides secure, scalable, and reliable data storage

Power BI

- Used as the analytics and visualization tool
- Connects directly to Azure SQL Database
- Enables real-time data analysis and dashboard creation

3. Data Loading and Preprocessing

The dataset is loaded into Azure SQL Database and connected to Power BI for analysis.

During preprocessing in Power BI:

- Missing values are handled
- Data types are validated
- Duplicate records are removed
- Data is structured into proper relationships

This ensures that the dataset is clean and ready for analysis.

4. Performance Analysis and Processing

Once the data is prepared, analytical operations are performed using Power BI and DAX.

Student Performance Analysis

- Subject-wise marks comparison
- Overall student performance evaluation
- Identification of high and low performers

Attendance Analysis

- Relationship between attendance and marks
- Identification of students with low attendance

Trend Analysis

- Average marks across subjects
- Pass and fail distribution

5. KPI Generation using DAX

Key Performance Indicators (KPIs) are created using DAX (Data Analysis Expressions):

- Total Students
- Average Marks
- Pass Percentage
- Fail Count

These KPIs provide a quick summary of student performance and help in decision-making.

6. Visualization and Dashboard Creation

The results are visualized using Power BI dashboards with:

- Bar charts for subject-wise performance
- Pie charts for pass/fail distribution
- KPI cards for quick insights
- Line charts for performance trends

These visualizations make it easier to understand patterns and insights.

7. Output and Performance Evaluation

The system outputs:

- Student performance summaries
- KPI-based insights
- Interactive dashboards
- Subject-wise and attendance-based analysis

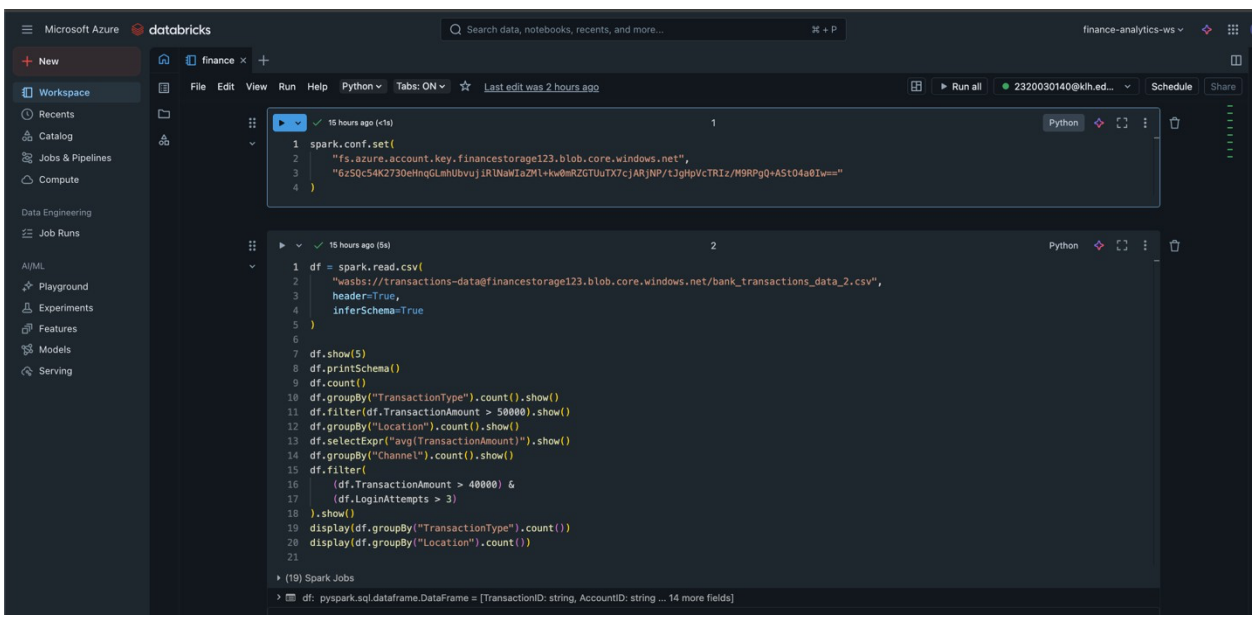
The performance of the system demonstrates that cloud-based analytics improves efficiency, scalability, and accuracy compared to traditional methods. It enables educators to monitor student progress effectively and make data-driven decisions.

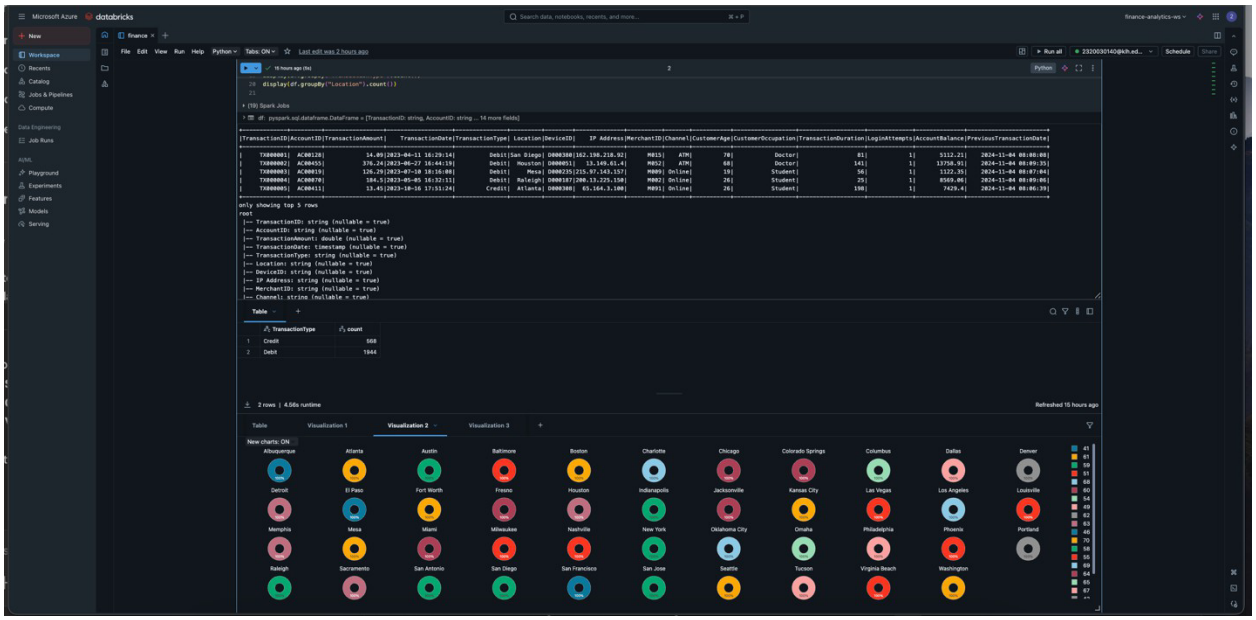
EXPEREMENTATION AND CODE:

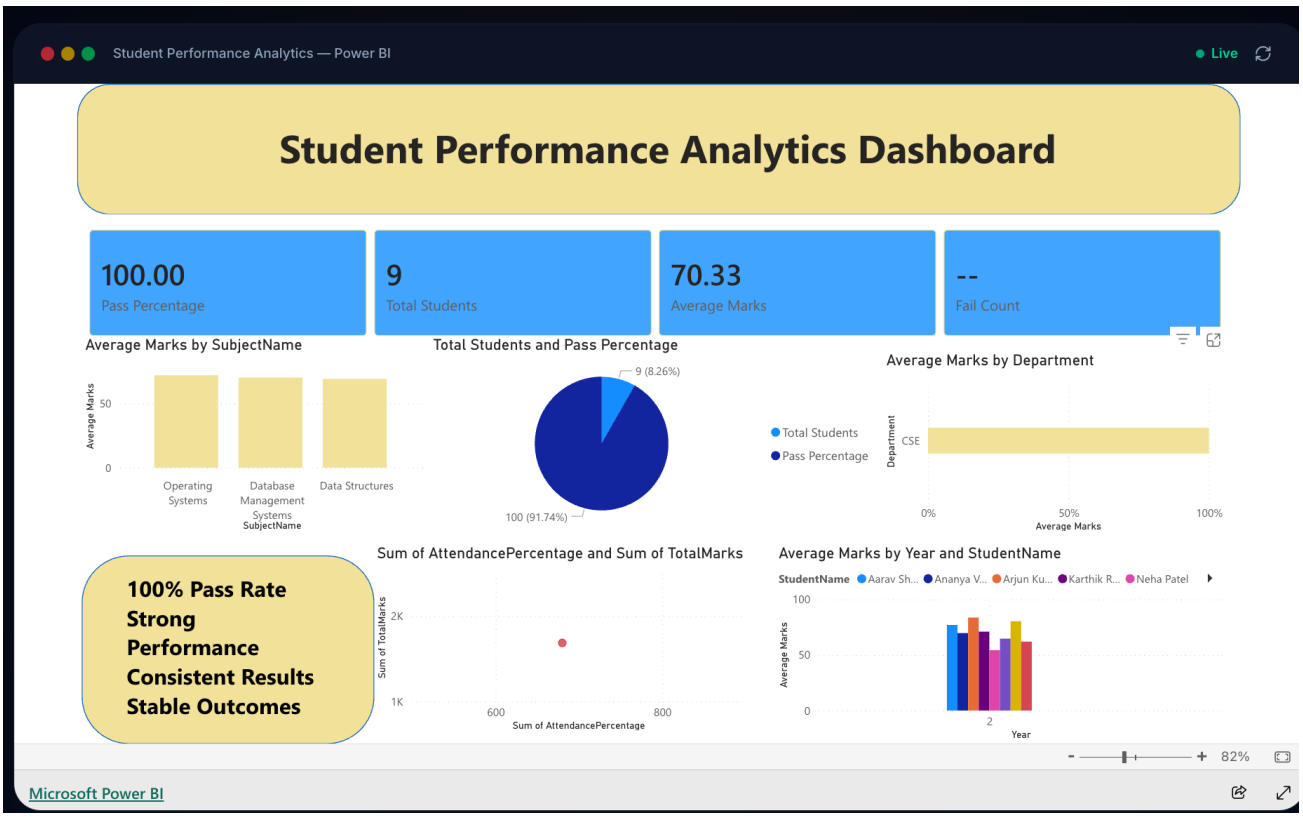
Code Structure

The project was implemented using Python in **Azure Databricks**. The code consists of the following main parts:

- **Data Loading:** Transaction data is loaded from Azure Blob Storage into Spark Data Frames.
- **Data Preprocessing:** Cleaning and formatting of transaction records.
- **Analysis Module:** Performs transaction analysis such as debit/credit count, location-wise analysis, and channel usage.
- **Fraud Detection Logic:** Identifies suspicious transactions using transaction amount and login attempts.
- **Visualization:** Bar charts and tables are used to display results.




Microsoft Azure
databricks
finance
File Edit View Run Help
Workspace
Recent
Catalog
Jobs & Pipelines
Compute
Job Runs
Playground
Experiments
Features
Models
Serving
display(df.groupBy("Location").count())
(10) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [TransactionID: string, AccountID: string ... 14 more fields]
only showing top 5 rows
root
|-- TransactionID: string (nullable = true)
|-- AccountID: string (nullable = true)
|-- TransactionAmount: double (nullable = true)
|-- TransactionDate: timestamp (nullable = true)
|-- TransactionType: string (nullable = true)
|-- Location: string (nullable = true)
|-- DeviceID: string (nullable = true)
|-- IP Address: string (nullable = true)
|-- MerchantID: string (nullable = true)
|-- Channel: string (nullable = true)
TransactionType
count
Credit
568
Debit
1944
2 rows | 4.56s runtime
Refreshed 15 hours ago
Visualization 1
Visualization 2
Visualization 3
New charts: ON


Student Performance Analytics — Power BI
Live
Student Performance Analytics Dashboard
100.00
Pass Percentage
9
Total Students
70.33
Average Marks
--
Fail Count
Average Marks by SubjectName
Average Marks
50
0
Operating Systems
Database Management Systems
Data Structures
SubjectName
Total Students and Pass Percentage
9 (8.26%)
100 (91.74%)
Average Marks by Department
Total Students
Pass Percentage
Department
CSE
0%
50%
100%
Average Marks
100% Pass Rate
Strong Performance
Consistent Results
Stable Outcomes
Sum of AttendancePercentage and Sum of TotalMarks
Sum of TotalMarks
2K
1K
600
800
Sum of AttendancePercentage
Average Marks by Year and StudentName
StudentName
Aarav Sh...
Ananya V...
Arjun Ku...
Karthik R...
Neha Patel
100
50
0
Average Marks
2
Year
82%
Microsoft Power BI

Experimentation

Steps Performed

1. Loaded financial transaction dataset into Azure Databricks.
2. Performed data cleaning and preprocessing.
3. Analysed transaction patterns such as:
    - Debit vs Credit transactions
    - Location-wise transactions
    - Channel-based usage
4. Detected high-value and suspicious transactions.
5. Applied a basic machine learning model (Logistic Regression) for fraud detection.
6. Visualized results using charts.

Performance Analysis

- **Processing Speed:** Fast due to cloud-based Spark processing
- **Scalability:** Handled large datasets efficiently
- **Accuracy:** Successfully identified suspicious transactions
- **Resource Usage:** Minimal local resource usage due to cloud execution

Conclusion of Experimentation

The experimentation proved that cloud-based analytics using Azure Databricks is efficient and reliable for financial transaction analysis. The system successfully identified transaction patterns and suspicious activities. The results show that cloud computing provides better scalability, performance, and flexibility compared to traditional systems.

RESULT

Results of the Student Performance Analytics Dashboard

The implementation of the *Student Performance Analytics Dashboard* produced meaningful results that demonstrate the effectiveness of cloud-based analytics in handling and analysing academic data. The project successfully analysed student records using Microsoft Azure SQL Database and Power BI, enabling efficient data processing, performance evaluation, and visualization of insights. The results highlight how cloud-based analytics can help educators monitor student progress, identify trends, and support academic decision-making.

1. Student Performance Analysis

The system analysed student data to understand overall academic performance. The following observations were obtained:

- Average student performance was around a moderate level, indicating balanced academic outcomes
- Subject-wise analysis revealed variations in performance, helping identify weaker subjects
- Some students consistently performed well, while others required additional academic support

These results demonstrate how analytics can help in evaluating student performance effectively.

2. Attendance and Performance Relationship

One of the key objectives of the project was to analyse the impact of attendance on student performance. The system observed:

- Students with higher attendance generally achieved better marks
- Low attendance was often associated with poor academic performance
- Attendance tracking helps identify students at risk

This analysis shows the importance of attendance in improving academic outcomes.

3. KPI-Based Performance Evaluation

Key Performance Indicators (KPIs) provided a quick summary of the system:

- Total number of students analysed
- Average marks across all subjects
- Pass percentage indicating overall success rate
- Fail count highlighting students needing improvement

These KPIs help educators quickly understand overall performance without analysing raw data.

4. Visualization and Insight Generation

Visualization played a crucial role in understanding the results. Power BI dashboards included:

- Bar charts for subject-wise performance comparison
- Pie charts for pass/fail distribution
- KPI cards for quick performance overview
- Trend graphs showing performance patterns

These visualizations made it easier to interpret complex academic data and enabled faster decision-making.

5. Performance Evaluation

Efficiency:
The use of cloud-based storage and Power BI enabled faster data processing compared to traditional methods.

Scalability:
The system can handle increasing student data without performance degradation, making it suitable for large institutions.

6. Learning Outcomes

Through this project, the following key learnings were achieved:

- Understanding of cloud-based data storage using Azure
- Practical experience with Power BI dashboards and DAX
- Knowledge of student performance analysis techniques
- Hands-on experience with data visualization
- Understanding the importance of data-driven decision-making in education

Final Overview

The results clearly demonstrate that cloud-based analytics is an efficient and reliable approach for analysing student performance. The system successfully identified performance trends, analysed attendance impact, and presented insights through interactive dashboards. Overall, the project achieved its objectives and highlighted how cloud computing and data analytics can improve academic performance monitoring and decision-making in educational institutions.

CONCLUTION

The Student Performance Analytics Dashboard project successfully demonstrated how cloud computing can be effectively used to analyze large volumes of academic data. By utilizing Microsoft Azure SQL Database and Power BI, the project showcased a scalable and efficient approach to storing, processing, and analyzing student performance data. The system was able to extract meaningful insights related to student performance trends, subject-wise analysis, and the impact of attendance on academic outcomes.

Through the implementation of data analytics techniques, the project highlighted how student data can be analyzed based on subjects, marks, attendance, and overall performance. The identification of key performance indicators such as average marks, pass percentage, and fail count demonstrated the importance of analytics in academic evaluation and progress tracking. The use of interactive dashboards and visualizations further enhanced the understanding of complex data by presenting it in an intuitive and user-friendly manner.

The integration of analytical measures using DAX added intelligence to the system by enabling dynamic calculations and real-time insights. This showed how cloud-based platforms can support advanced analytics and reporting capabilities with minimal infrastructure requirements.

Overall, the project successfully met its objectives by demonstrating the practical application of cloud computing in educational analytics. It provided valuable insights into how institutions can leverage cloud technologies for scalable data processing, improved decision-making, and better academic performance monitoring. The project also emphasized the importance of cloud-based solutions in handling real-world educational data efficiently and reliably.

This work serves as a strong foundation for future enhancements such as real-time data updates, predictive analysis using machine learning models, and expansion to multiple departments or institutions, making it highly relevant for modern education systems.

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